

Multiple Choice (10 marks)

1. The number of test examples needed to get statistically significant results should be __
 - (a) Larger if the error rate is larger.
 - (b) Larger if the error rate is smaller.
 - (c) Smaller if the error rate is smaller.
 - (d) It does not matter.
2. Suppose your model is overfitting. Which of the following is NOT a valid way to try and reduce the overfitting?
 - (a) Increase the amount of training data.
 - (b) Improve the optimisation algorithm being used for error minimisation.
 - (c) Decrease the model complexity.
 - (d) Reduce the noise in the training data.
3. Which image data augmentation is most common for natural images?
 - (a) random crop and horizontal flip
 - (b) random crop and vertical flip
 - (c) posterization
 - (d) dithering
4. For polynomial regression, which one of these structural assumptions is the one that most affects the trade-off between underfitting and overfitting:
 - (a) The polynomial degree
 - (b) Whether we learn the weights by matrix inversion or gradient descent
 - (c) The assumed variance of the Gaussian noise
 - (d) The use of a constant-term unit input
5. The disadvantage of Grid search is
 - (a) It can not be applied to non-differentiable functions.
 - (b) It can not be applied to non-continuous functions.
 - (c) It is hard to implement.
 - (d) It runs reasonably slow for multiple linear regression.

True / False (10 marks)

Answer True or False

6. True or False: The joint probability distribution $P(X, Y, Z)$ can be expressed as $P(Y) * P(X|Y) * P(Z|Y)$ in the given Bayes net.
7. True or False: Hidden Markov models are primarily used for classifying images rather than modeling sequences like English sentences.
8. True or False: Word 2 Vec parameters are typically not initialized using a Restricted Boltzmann Machine.
9. True or False: The original ResNets were primarily trained using the SGD optimizer for all experiments.
10. True or False: Adding more basis functions in a linear model decreases variance, leading to a more stable model across different datasets.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to find the first repeated word in a given string.
12. Write a function to count those characters which have vowels as their neighbors in the given string.

End of Paper